

Lab 2

Introduction to Status Flags

Prof. Charan Bhaskar

Register groups

To reduce clutter, the **Registers** window organizes registers into groups. If you right-click on the **Registers** window, you will see a shortcut menu containing a list of groups, which you can display or hide as you see fit.

Register flags

For Intel x86 processors, you may see the following flags in the **Registers** window. There are two types of flags: Status Flags and Control Flags.

In this lab, we will deal with status flags.

Status Flags

1. Sign (PL):

After any operation if the MSB (7th bit) of the result is 1, it indicates the number is negative and the sign flag becomes set, i.e. 1. If the MSB is 0, it indicates the number is positive and the sign flag becomes reset i.e. 0.

From 00H to 7F, sign flag is 0

From 80H to FF, sign flag is 1

1- MSB is 1 (negative)

0- MSB is 0 (positive)

```
MOV AL, 10 (10 is 0000 1010 which is positive)
```

```
MOV BL, 9 (32 is 0000 1001 which is positive)
```

```
SUB BL, AL (-1 is 1111 1111 which is negative)
```

2. Overflow (OV):

This flag will be set (1) if the result of a signed operation is too large to fit in the number of bits available to represent it, otherwise reset (0).

After any operation, if D[6] generates any carry and passes to D[7] OR if D[6] does not generate carry but D[7] generates, overflow flag becomes set, i.e., 1.

If D[6] and D[7] both generate carry or both do not generate any carry, then overflow flag becomes reset, i.e., 0.

Example: On adding bytes 100 + 50 (result is not in range -128...127), so overflow flag will set.

```
MOV AL, 100
MOV BL, 50
ADD AL, BL
```

3. Zero (ZR):

After any arithmetical or logical operation if the result is 0 (00)H, the zero flag becomes set i.e. 1, otherwise it becomes reset.

Example:

```
mov al, 50
mov bl, 50
cmp al,bl    ;ZR=1
sub al, bl
cmp al,bl    ;ZR=0
```

4. Auxiliary Carry (AC):

If after any arithmetic or logical operation D(3) generates any carry and passes it on to D(4) this flag becomes set i.e. 1, otherwise, it becomes reset i.e. 0.

Example: These set of instructions will set the auxiliary carry flag to 1, as on adding 2B and 39, the addition of lower-order nibbles B and 9 will generate a carry.

```
mov al, 39h ; 39h is 0011 1001
mov bl, 2bh ; 2Bh is 0010 1011
add al, bl   ; sum is 0110 0100
```

5. Parity (PE):

If after any arithmetic or logical operation the result has even parity, an even number of 1 bit, the parity register becomes set i.e. 1, otherwise it becomes reset i.e. 0.

Example:

```
mov al, 1 ; 0000 0001 has odd number of 1
mov bl, 1 ; 0000 0001 has odd number of 1
add bl, al ; 0000 0010 has odd number of 1
add bl,al ; 0000 0011 has even number of 1
```

6. Carry (CY):

Carry is generated when performing n bit operations and the result is more than n bits, then this flag becomes set i.e. 1, otherwise, it becomes reset i.e. 0.

During subtraction (A-B), if $A > B$ it becomes reset, and if $(A < B)$ it becomes set.

Carry flag is also called the **borrow flag**.

1-carry out from MSB bit on addition or borrow into MSB bit on subtraction

0-no carry out or borrow into MSB bit.

Example:

```
MOV AL, 0FFh (1111 1111)
MOV BL, 1h   (0000 0001)
ADD AL, BL   (0000 0000)
```

```
MOV AL, 10 (10 is 0000 1010)
MOV BL, 9  (32 is 0000 1001)
SUB BL, AL (-1 is 1111 1111)
```

Control Flags

The control flags enable or disable certain operations of the microprocessor.

7. Directional Flag (DF)

This flag is specifically used in string instructions. If directional flag is set (1), then access the string data from higher memory location towards lower memory location. If directional flag is reset (0), then access the string data from lower memory location towards higher memory location.

8. Interrupt Flag (IF)

This flag is for interrupts. If interrupt flag is set (1), the microprocessor will recognize interrupt requests from the peripherals. If interrupt flag is reset (0), the microprocessor will not recognize any interrupt requests and will ignore them.

Lab Work

- Execute sample code given for Status Flags 1 to 6 in the manual.
- For each sample code, create a table as below and fill the values at each stage.

Steps	Instruction	AX	BX	OV	PL	ZR	AC	PE	CY
1	MOV AL, 10								
2									
...									

- Explain why flag values are getting changed for each sample code.